

Business Recommendations and Insights

The evaluation also establishes HbA1c level, blood glucose level, BMI, and age as the most powerful predictors of the presence of diabetes. HbA1c and glucose indicate a strong correlation with diabetes, whereas the higher the BMI and the older the age, the more exposed to risk the individual is in the long term. These predictors are especially useful in the business view as they are regularly gathered in regular medical checkups and hence, the model is viable to be put to practical use in the real world without extra expenses associated with the collection of data.

The results of the model can be used by healthcare organizations, insurance providers, and digital health platforms to make decisions on preventing health issues. The predictions can be used by the hospitals to put high-risk patients at the top of the list in order to screen them early, counsel them on nutrition, and provide them with follow-up tests. Insurance companies are able to develop preventive care schemes to the determined high-risk people which lowers the cost of claims in the future. Likewise, model outputs can be used to provide telemedicine platforms with the recommendations to prescribe diagnostic tests and lifestyle interventions at an early stage of the disease, thus increasing patient outcomes and operational efficiency.

Early diagnosis saves a lot of money in healthcare expenses through avoiding late-stage complications that are costly like cardiovascular diseases, kidney diseases, and long hospitalization. In this regard, as an example, the high-risk or pre-diabetic patients can be identified and low-cost interventions can be implemented like lifestyle modification and regular check-ups are much cheaper than medication or surgical intervention in the long term. This proactive strategy will reduce hospitalizations and decrease healthcare expenditures and insurance costs.

Precision and recall are specifically significant when it comes to healthcare applications. Whereas high precision reduces the unjustified care of healthy patients, high recall is more important due to the complicating impact of being a diabetic patient (false negative) and damaging the health system. Missing a high-risk patient in medical decision-making is a more dangerous problem than providing more screenings to low-risk patients, so recall is a highly important measure.

The model has some limitations even though it has a good performance. In the dataset, there is an imbalance in classes, and there are much more cases of diabetic patients that can bias the prediction to the majority group. Also, the SMOTE techniques were not used in oversampling, which could potentially result in the inability of the model to accurately recognize minority-class patients. There may also be the presence of data bias because of the concentration of the data at a demographic level, which diminishes the ability to extrapolate the data to different populations.

In the future, it may be improved by using a class-weighted learning or SMOTE to increase the recall of diabetic cases. The data could be extended to serve more diverse population to enhance the power of the model. Moreover, involving the model in the real-time healthcare framework and retraining it on a continuous basis with the additional patient data would have improved the predictive accuracy and long-run business value.